

CSS Codes

Code Space of CSS codes

$z_1, \dots, z_s \in \Sigma^n$ $\rightarrow$ n -bit parity checks corresponding to Z -sg

$$C_z = \{u \in \Sigma^n : u \cdot z_1 = \dots = u \cdot z_s = 0\}$$

$x_1, \dots, x_t \in \Sigma^n$ $\rightarrow$ n -bit parity checks corresponding to X -sg

$$C_x = \{u \in \Sigma^n : u \cdot x_1 = \dots = u \cdot x_t = 0\}$$

$$\left. \begin{aligned} C_z^\perp &= \{z_1 z_1 \oplus \dots \oplus \alpha_s z_s : \{\alpha_1, \dots, \alpha_s\} \in \{0, 1\}^s\} \\ C_x^\perp &= \{z_1 x_1 \oplus \dots \oplus \alpha_t x_t : \{\alpha_1, \dots, \alpha_t\} \in \{0, 1\}^t\} \end{aligned} \right\} \text{Dual Codes}$$

$$C_2^h \subseteq C_x \quad C_x^h \subseteq C_2$$

Code space is spanned by vectors of the form

$$|u \oplus C_x^h\rangle = \frac{1}{\sqrt{2^t}} \sum_{v \in C_x^h} |u \oplus v\rangle \quad \text{for all } u \in C_2$$

↳ These vectors are uniform super position over the strings in dual code C_x^h shifted by / XORed with strings in the

code C_2

To explain it, say Π is the projection onto the code space

→ We can now swap X and Z too, to swap $\{|0\rangle, |1\rangle\}$ with $\{|+\rangle, |-\rangle\}$ we apply $H^{\otimes n}$

$$H^{\otimes n} |u \oplus C_2^h\rangle = \frac{1}{\sqrt{2^n}} \sum_{v \in C_2^h} H^{\otimes n} |u \oplus v\rangle \text{ for all } u \in C_x$$

→ Ex. → 7-qubit steane code

